

Cooperativity in Enzyme Kinetics

How a single enzyme turns a gradual change in substrate into a sharp, switch-like change in rate — and how effectors move that switch.

Describe it

Hill equation, Hill plot, the 81-fold rule.

Derive it

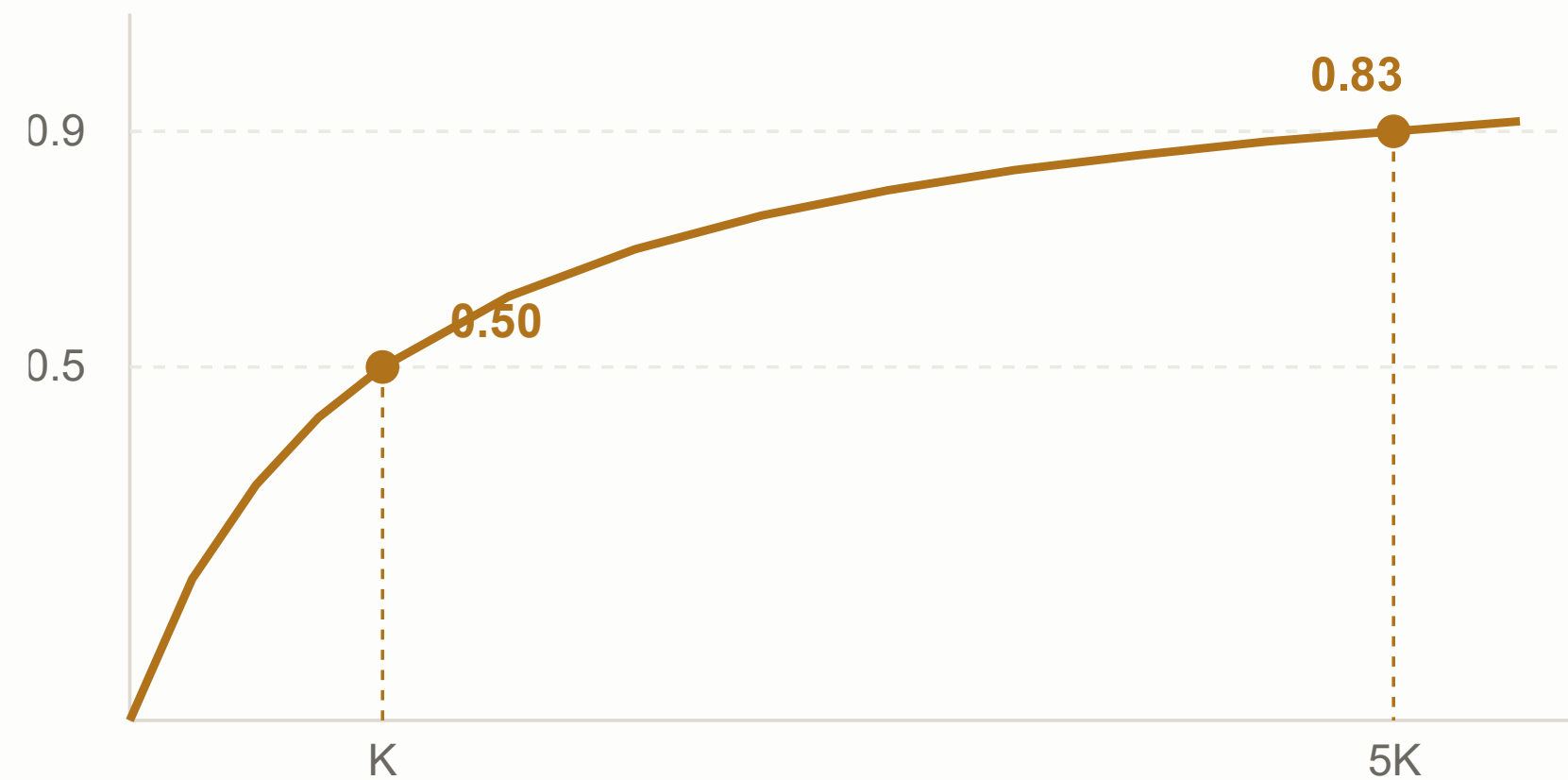
Adair binding constants; the MWC and KNF models.

Use it

Haemoglobin, ATCase, PFK, and lactic acid bacteria.

WHERE WE START

The Limits of Michaelis–Menten Kinetics



$r/r_{\max}$ against substrate concentration s

$$r = r_{\max} \frac{s}{s + K}$$

A hyperbola rises steeply at low substrate and then flattens. There is no concentration range in which the rate the rate changes sharply.

$$s : K \rightarrow 5K \Rightarrow r/r_{\max} : 0.50 \rightarrow 0.83$$

Five-fold on the input, 1.7-fold on the output. And to cross $0.1 \rightarrow 0.9$ requires an 81-fold rise in s — for *every* hyperbola, whatever K may be.

EMPIRICAL DESCRIPTION

The Hill Equation

$$r = r_{\max} \frac{s^n}{s^n + K_n}$$

EQUIVALENTLY

$$\theta = s^n / (s^n + K_n)$$

θ = fractional saturation of the sites

$n > 1$

Positive cooperativity — one bound molecule makes the next bind more easily.

$n < 1$

Negative cooperativity — each bound molecule hinders the next.

$n = 1$

Michaelis–Menten exactly. The Hill form generalises it rather than replacing it.

$\theta = 0.5$ at $s = K$ for every n . Hill (1910), oxygen binding to haemoglobin. n is fitted, not counted: four sites, $n \approx 2.8$.
 n .

EMPIRICAL DESCRIPTION

The Hill Plot

REARRANGE, THEN TAKE LOGS

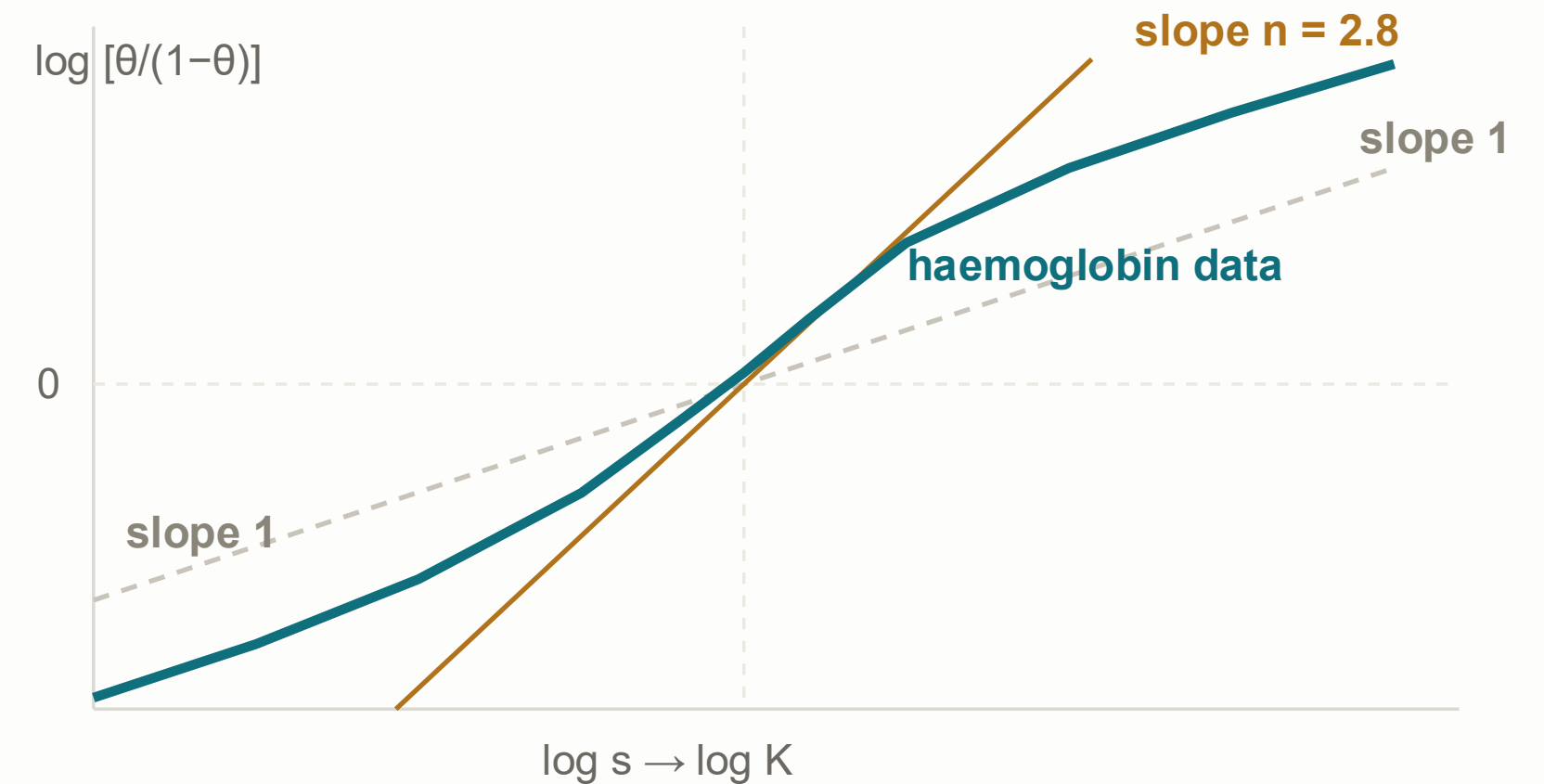
$$\frac{\theta}{1 - \theta} = \frac{s^n}{K^n}$$

$$\log [\theta / (1 - \theta)] = n \log s - n \log K$$

A straight line of slope n crossing zero at $s = K$. This is how a Hill coefficient is actually obtained from data.

Real data are not straight. Haemoglobin gives slope ≈ 1 at both extremes and ≈ 2.8 only in mid-saturation, so the fitted n is a *maximum* slope, not a constant.

At very low θ only one site is occupied; at very high θ only one is free — free — neither limit has anything to cooperate with.



EMPIRICAL DESCRIPTION

Steepness: the 81-Fold Rule

Take the ratio of the substrate concentrations giving $\theta =$ giving $\theta = 0.9$ and $\theta = 0.1$. Every K cancels:

$$\frac{s_{0.9}}{s_{0.1}} = 81^{1/n}$$

The whole working range of the enzyme is compressed by the n -th root of 81. Nothing else about the enzyme enters.

Same rule at the other common pair: for $n = 4$, moving θ $0.5 \rightarrow 0.9$ takes $s : K \rightarrow 1.73K$, against $K \rightarrow 9K$ for a hyperbola.

Hill coefficient n	$s_{0.9} / s_{0.1}$	Character
1	81×	Hyperbolic — no useful threshold
2	9.0×	Clearly sigmoidal
2.8	4.4×	Haemoglobin
4	3.0×	Switch-like
10	1.6×	Near step function

DERIVATION

Sequential Binding: the Adair Equation

Give each binding step its own association constant rather than one lumped exponent. For two sites, $E + s \rightleftharpoons Es$ (K_1)
sites, $E + s \rightleftharpoons Es$ (K_1) and $Es + s \rightleftharpoons Es_2$ (K_2):

$$\theta = \frac{K_1s + 2K_1K_2s^2}{2(1 + K_1s + K_1K_2s^2)}$$

THE FACTORS OF 2

Statistical, not energetic. There are two ways to fill one of two sites, one way to fill both.

COOPERATIVITY CRITERION

$$4K_2 > K_1$$

Positive cooperativity once the statistical factor is removed; equality gives a hyperbola.

FOUR SITES

Same construction with $K_1...K_4$. For For haemoglobin the fourth O_2 binds binds roughly **100×** more tightly than than the first.

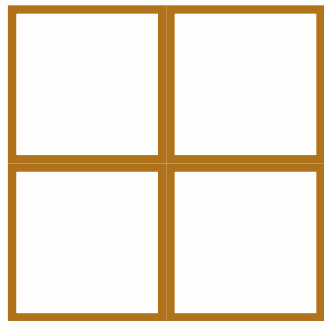
Adair describes; it does not explain. explain.

It tells us the constants differ, not what makes them differ. That question needs a structural model.

DERIVATION

The Concerted Model (MWC)

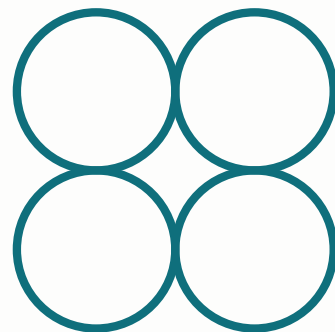
T · low affinity



$$L = [T_0]/[R_0]$$



R · high affinity



all subunits switch together — no mixed states

$\alpha = s / K_R$ scaled substrate

$c = K_R / K_T$ affinity ratio, $0 \leq c < 1$

L conformational equilibrium with no ligand bound

SATURATION FUNCTION FOR N SITES

$$Y = \frac{Lc\alpha(1 + c\alpha)^{n-1} + \alpha(1 + \alpha)^{n-1}}{L(1 + c\alpha)^n + (1 + \alpha)^n}$$

Substrate binds R preferentially, so each binding event pulls the equilibrium toward R and toward R and raises the affinity of the remaining sites. The sigmoid is a by-product of a product of a shifting conformational equilibrium.

STRONG
SIGMOID

large L, small c — the enzyme starts firmly in T and R binds far better

NO SIGMOID

$L \rightarrow 0$ or $c \rightarrow 1$ — the curve collapses to a hyperbola

EFFECTORS

activators lower L, inhibitors raise L — the switch moves without touching the active site

The model cannot produce negative cooperativity at all — a structural prediction, and a prediction, and a testable one.

DERIVATION

The Sequential Model (KNF)



WHAT CHANGES

Each ligand induces a change in *its own* subunit, which alters neighbouring affinities through the subunit interfaces. Mixed conformational states are permitted.

If the induced change makes neighbours bind *worse*, the fitted n drops below 1 — negative cooperativity, which MWC forbids.

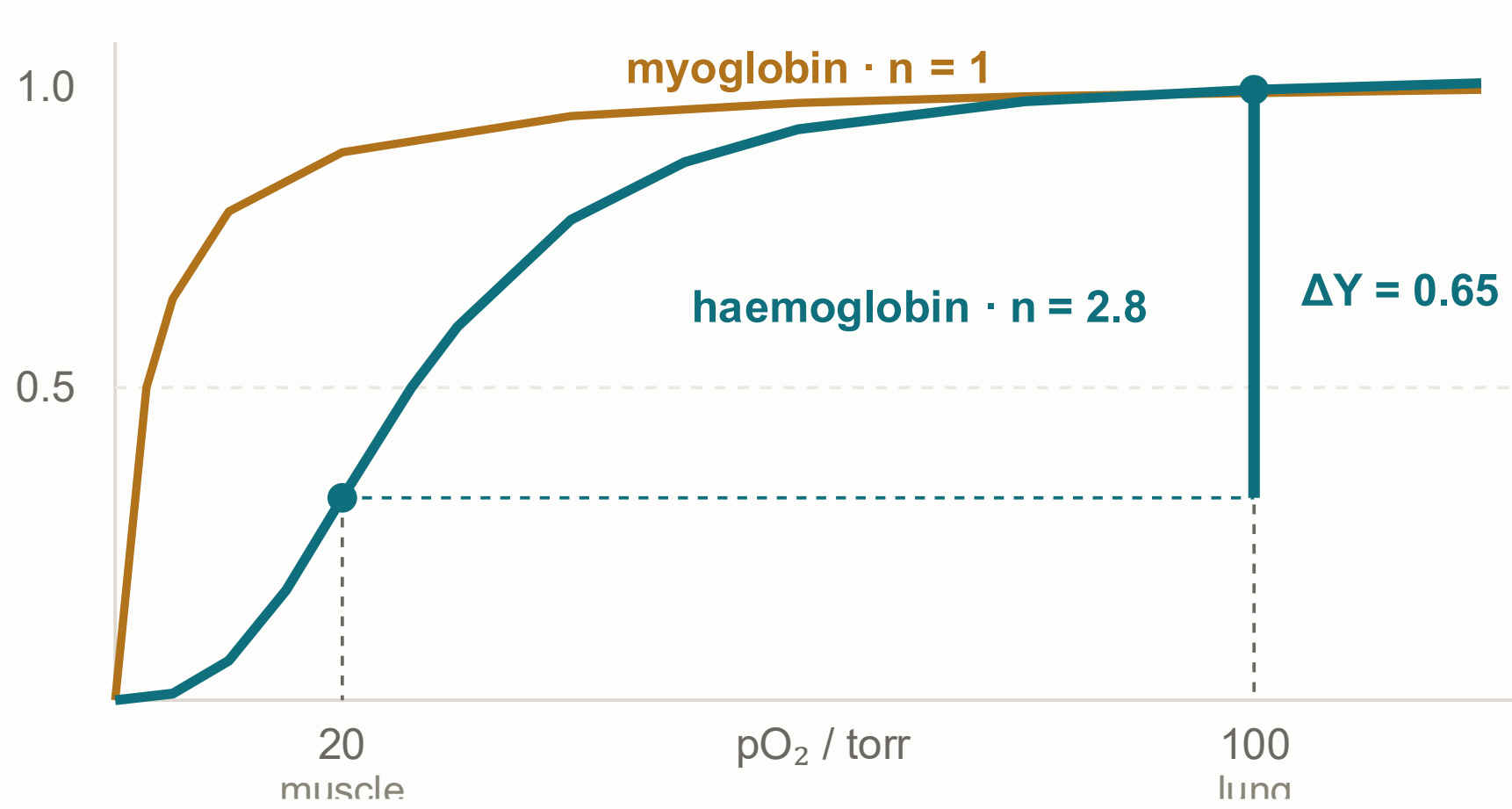
WHICH MODEL WINS

Neither, universally. Haemoglobin is well described by MWC; glyceraldehyde-3-phosphate dehydrogenase shows negative cooperativity that only KNF can accommodate.

Both reduce to a Hill-type curve over the measurable range, which is why the empirical n remains the working currency.

WORKED EXAMPLE

Haemoglobin: Oxygen Delivery



	P ₅₀	n	Y at 100	Y at 20
Haemoglobin	26	2.8	0.98	0.32
Same P ₅₀ , n = 1	26	1.0	0.79	0.43

0.65 vs 0.36

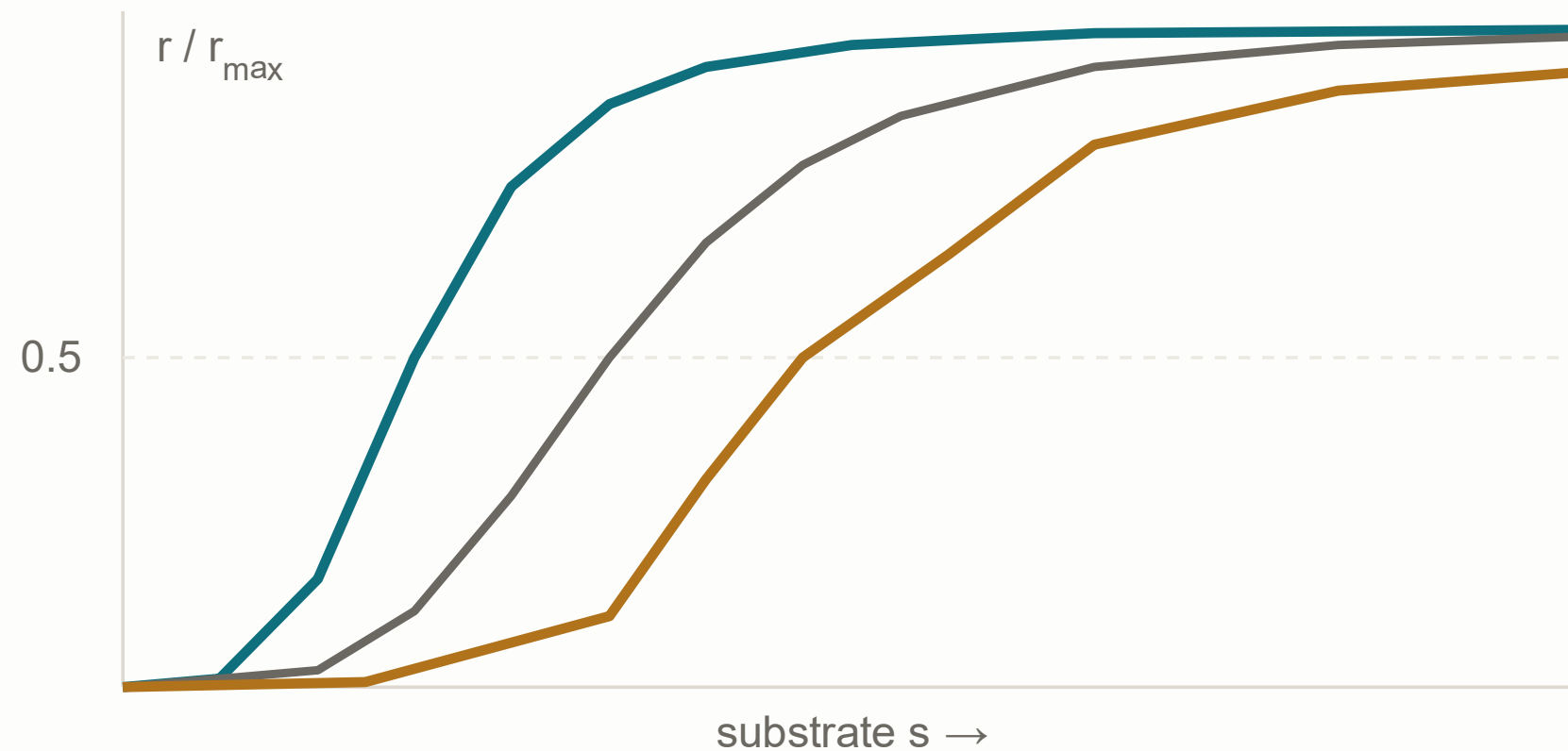
Fraction of capacity unloaded per circuit. Cooperativity nearly doubles oxygen delivery without changing the affinity midpoint at all.

EFFECTORS SHIFT THE CURVE

2,3-bisphosphoglycerate and H⁺ stabilise the T state, raising L and L and P₅₀ — the curve moves right and more O₂ is released in acidic, in acidic, working tissue (Bohr effect).

REGULATION

Allosteric Effectors: K-Systems and V-Systems



— + activator — no effector — + inhibitor
activator effector inhibitor

A K-system: the effector slides the sigmoid sideways, $r_{\max}$ unchanged.

Cooperative binding need not involve the substrate. An activator or inhibitor at a separate regulatory site does the same job — the definition of allosteric regulation.

K-SYSTEM · APPARENT K MOVES

ATCase — CTP, the end product, raises apparent K; ATP lowers it.

PFK-1 — ATP and citrate inhibit; AMP and fructose-2,6-bisphosphate activate.

V-SYSTEM · $R_{\max}$ MOVES

The effector changes the catalytic rate rather than the affinity; the curve rises or falls in place.

In MWC terms both are one parameter change: activators lower L , inhibitors raise it. A small shift in effector concentration produces a large flux change precisely because the curve is steep.

REGULATION

Feed-Forward Activation in Lactic Acid Bacteria

Fructose-1,6- diphosphate

accumulates when glycolytic flux is high



ACTIVATES

INHIBITS

lactate dehydrogenase
dehydrogenase
pyruvate formate lyase
lyase

High flux therefore pushes pyruvate toward lactate automatically — a feed-forward signal carried by an intermediate, intermediate, with no dedicated sensor.

Excess sugar

High FDP, homolactic fermentation — essentially all carbon to lactate.

Sugar limitation

Low FDP, mixed-acid fermentation — formate, acetate and ethanol appear.

SUMMARY

Cooperativity at a Glance

Description	Parameters	Answers	Limitation
Hill	n, K	How steep is the response?	n is empirical and varies with saturation
Adair	$K_1 \dots K_n$	Which binding step changes affinity?	Describes without explaining; many parameters
MWC	L, c, n	Why does affinity change, and how do effectors act?	Cannot give negative cooperativity
KNF	Per-subunit interaction constants	What if subunits change one at a time?	Parameter-heavy, harder to falsify

Two numbers to keep: $s_{0.9}/s_{0.1} = 81^{1/n}$ and haemoglobin's 0.65 against 0.36 unloaded.